Project Proposal – Scholarly Wikidata

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Quickest way to add data to Wikidata!

Project idea

I propose a small web application that helps researchers turn information about musical keyboards into facts for a knowledge graph. A researcher will upload a PDF or text file, paste text, or provide a website link. The application will read the source, suggest facts, and match them to the items and properties used by the knowledge base. The researcher will review each suggestion before approving it for addition.

Inspired by Scholarly Wikidata, this project will build on the existing Keyboards Wikibase created by Professor Finn Årup Nielsen. The proof of concept will use a local copy of this knowledge graph to test adding reviewed facts. It will focus on musical keyboards, including synthesizers with keys, digital pianos and organs, with later support for preparing Wikidata statements. This project will also serve as a basis, maybe even be turned into a tool for companies to create their own Knowledge Bases.

Why this is useful

Specifications are spread across manuals, product pages and catalogues. Moving this information into a knowledge graph requires reading the source, identifying the correct model, choosing a property and checking the value. This project aims to reduce that repetitive work while keeping the researcher responsible for deciding which facts are reliable.

How the application will work

The application will first extract readable text. For scanned pages, it will use optical character recognition, or OCR, to turn images of text into words. It will keep a link to the source and, where available, the page number and original passage.

It will then identify facts and arrange them as subject–predicate–object triples. For example, “Yamaha manufactured the Yamaha DX7” becomes “Yamaha DX7 — manufacturer — Yamaha”. Here, the subject is the instrument, the predicate is the relationship, and the object is its value.

Next, the application will suggest matching item identifiers, called QIDs, and property identifiers, called PIDs. An object that names an organisation or instrument can also need an item identifier. Values such as weight or dates will keep their units or date precision. Matching will use the existing keyboard graph's identifiers, which can differ from Wikidata identifiers.

The review screen will show each proposed triple beside its supporting passage and suggested matches. The researcher can correct, approve or reject it. Approved facts will be added to the local graph together with their source information.

Implementation plan

The work will build on the existing text extraction and Wikidata lookup prototype. That prototype already produces triples, searches for subject and property identifiers, and uses pauses and caching to reduce repeated API requests. The next steps will connect these parts to a reviewable web workflow.

1 Use the existing keyboard graph. Inspect the items and properties in Professor Nielsen's Keyboards Wikibase and prepare a local copy for testing. Select an initial set of existing properties and their allowed value types, map them to Wikidata where suitable, and prepare a small set of checked examples.

2 Add document and website input. Build an upload form and website-link field. Extract text from PDFs, text files and accessible web pages, and add OCR for scanned PDFs. Preserve source passages and page references so that researchers can verify suggestions.

3 Improve extraction and matching. Keep similar keyboard models and variants separate. Compare several possible item matches using their names, descriptions and the source passage. Flag uncertain or missing matches for review. Use the agreed property list to reduce failures caused by different names for the same property.

4 Build review and submission. Show proposed facts in an editable review table with their sources. Allow approval and rejection per fact. Check for existing statements before saving, report conflicts and show which approved facts were successfully added to the local graph.

5 Test the complete workflow. Try readable PDFs, scanned pages, pasted text and websites. Include sources that mention several keyboard models and accessories. Compare the suggestions with manually checked facts, then use the errors to improve the application.

How success will be measured

Evaluation will measure how many suggested facts are correct, how many expected facts are found, and how often the item and property matches are correct. It will also record processing time and the corrections needed during review. A successful demonstration will take a source through extraction, review and approval, then show the approved facts and their sources in the local graph.

Main challenges and expected outcome

Poor scans, crowded specification tables, similar model names and missing properties can all lead to mistakes. Source passages and explicit review will help researchers identify these errors. Uncertain facts will remain available for correction, and saving will require approval.

The expected outcome is a working web application, a small checked evaluation set, and a short report describing its accuracy and limitations. The proof of concept will demonstrate the full process using the local keyboard graph and provide a basis for later Wikidata integration.